

DSC Capstone Quarter 2 Proposal

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1 Intro

In California wildfires are an extremely common occurrence we face yearly. With such a destructive natural disaster that is so prevalent in the state, it is crucial that companies can correctly estimate the risk of property damage and the severity of the damage, which affects resource allocation.

1.1 Literature assessment

In the article “Unrefined national building inventories can mislead risk assessments and decisions” Adam B. Pollack et al ¹. discuss the importance of having accurate data in risk assessment models. The authors, who are all research scientists, wrote the paper to bring attention to how inaccurate data can impact risk assessment models. They compared the NSI (National Structure Inventory) property dataset with high-quality local data from Philadelphia, Pennsylvania. As an example, it was discovered that the NSI data matched the Philadelphia property on the 3 fundamental characteristics (location, presence of a basement, number of floors) for only 36% residential buildings (p. 9). The findings conclude that the NSI data without refinements can lead to distortion of damage estimates and by extension problems with resource allocation. The refinements that could significantly improve the results include fixing story and basement presence misclassifications, property location misplacements, and so on. The study suggests that completing even small low-cost refinements can significantly improve the assessment models.

1.2 Project Objective

While Pollack et al ². focus primarily on flood risk, the same issues apply to other natural disasters, namely our focus wildfires. In our project we will examine property data through the wildfire risk assessment lens and explore the following question: How can we detect erroneous property characteristic data, and correct data that is not consistent throughout the dataset? It is evident just how crucial spotting problems in data is, and we focus on effectively spotting the errors and improving the data quality.

2 Methods to Explore

2.1 Poisson regression for Expected Count Modeling

A method we will explore is Poisson regression, which is a generalized linear model used for modeling count data ³. We could model the sum of cc for each combination with pre-

¹Polack, F. (2023).

²Polack, F. (2023).

³Anderson, C. J. (n.d.).

dictors land use class, structure class, building type, number of stories, square footage bin, structural value bin. After we produce a model we can calculate expected counts and find cells where the count is much larger or smaller than the model's predicted count, these discrepancies will be used to flag potentially erroneous data.

2.2 Local Moran's I for Spatial Outlier Detection

Another method we will explore is Local Moran's I⁴, a spatial statistic used to identify clusters or outliers relative to neighboring areas. We will compute relative frequencies for property characteristics and compare each county to the average of its neighbors. Counties that are statistically inconsistent with their neighbors will be treated as spatial outliers and flagged as places where we should check the data.

2.3 Local Influence Function (LIF)

The Local Influence Function (LIF)⁵ measures how individual observations affect local spatial autocorrelation. Unlike Moran's I, which summarizes spatial patterns, LIF identifies observations that disproportionately distort those patterns. LIF accounts for both magnitude and spatial dependence, making it effective at detecting subtle or localized errors in structural attributes.

2.4 ODAR Density-Based Outlier Detection

The ODAR framework⁶ transforms the feature space so that normal records and outliers become separable. This enables clustering algorithms to identify outlier clusters directly. We will apply ODAR to detect structural characteristics that form abnormal clusters inconsistent with expected property distributions.

3 Hypothesis

3.1 Detecting Outliers

We will obtain statistics on property characteristics of structures in the NSI dataset. By ranking them, we will be able to identify outliers. Different methods of detecting outliers will likely have different results, which reflect potential strengths and weaknesses for each

⁴Li, T., Fernandez, A., & Yu, C. (2024)

⁵Li, T., Fernandez, A., & Yu, C. (2024)

⁶Li, Q., & Wang, S. (2024).

method. For example, using a method based on the mode of a distribution may not accurately summarize the entire county, and may only be an accurate reflection of the majority or plurality of areas within a county instead.

3.2 Correcting Data

Generally, we expect to see that more fine grained approaches will yield better results than more generalized ones. For example, a very rough method of correcting outliers would be to impute the mean or median for each county. A similar method was used in the article “Unrefined national building inventories can mislead risk assessments and decisions”. Adam B. Pollack et al. similarly impute the values in the categories they analyze using the median, but have better results imputing values that are instead aggregated over tracts, which are subregions that are smaller than counties, which is a more detailed method. In addition to using aggregated values, we can also attempt methods like K-nearest neighbors to replace outliers with values that are closer to other rows with similar attributes.

4 Timeline

Table 1: Seven-Week Project Timeline and Responsibilities

Team Member	W1	W2	W3	W4	W5	W6	W7
General Plan	Individually work + share findings	Evaluate prior models	Improve top 2 models + visualization	Evaluate best solution	Choose visualization approach	Website poster	Website poster
Tatiana Samokhvalova	Density-Based Outliers	Model improvements	Isolation Forest	Model improvements	Visualize (BQ/QGIS)	Website	Website
Angela Shen	Spatial residual modeling	Model improvements	Improve + visualize	Evaluate best	Visualize (BQ/QGIS)	Website	Website
Sardor Sobirov	Remote-sensing CV	Model improvements	Improve + visualize	Evaluate best	Visualize (BQ/QGIS)	Poster	Poster
Nathaphat Taleongpong	Poisson regression	Model improvements	Autoencoders	Model improvements	Visualize (BQ/QGIS)	Poster	Poster

5 Primary Output

The primary outputs of this project will be a website, a written report, and a poster. The core deliverable will be code that processes spatial data to generate maps and flag observations that appear suspicious. The data analyses will be communicated through static maps and figures in the report and poster. The website will feature an interactive web map (for example using GeoViz or Leaflet) that allows users to explore the data.

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